

Linear Algebra (MT-1004)

Homework #33

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Question 1

Consider the set of functions $S = \{e^x, e^{2x}, e^{3x}\}$ defined on the interval $I = [0, 1]$. Use the **Gram-Schmidt process** to orthogonalize this set with respect to the inner product:

$$\langle f, g \rangle = \int_0^1 f(x)g(x) dx.$$

Solution

Step 1: Define the Inner Product

The inner product is:

$$\langle f, g \rangle = \int_0^1 f(x)g(x) dx.$$

Step 2: Begin the Gram-Schmidt Process

Step 2.1: Compute v_1

Let $v_1 = e^x$. Normalize v_1 to obtain u_1 :

$$\begin{aligned}\|v_1\| &= \sqrt{\langle v_1, v_1 \rangle} = \sqrt{\int_0^1 (e^x)^2 dx} \\ \|v_1\| &= \sqrt{\int_0^1 e^{2x} dx} = \sqrt{\left[\frac{e^{2x}}{2}\right]_0^1} = \sqrt{\frac{e^2 - 1}{2}}. \\ u_1 &= \frac{v_1}{\|v_1\|} = \frac{e^x}{\sqrt{\frac{e^2 - 1}{2}}}.\end{aligned}$$

Step 2.2: Compute v_2

$$v_2 = e^{2x} - \text{proj}_{u_1}(e^{2x}),$$

where:

$$\text{proj}_{u_1}(e^{2x}) = \langle e^{2x}, u_1 \rangle u_1.$$

Calculate $\langle e^{2x}, u_1 \rangle$:

$$\langle e^{2x}, u_1 \rangle = \int_0^1 e^{2x} \cdot \frac{e^x}{\sqrt{\frac{e^2-1}{2}}} dx = \frac{1}{\sqrt{\frac{e^2-1}{2}}} \int_0^1 e^{3x} dx.$$

$$\int_0^1 e^{3x} dx = \left[\frac{e^{3x}}{3} \right]_0^1 = \frac{e^3 - 1}{3}.$$

Thus:

$$\langle e^{2x}, u_1 \rangle = \frac{\frac{e^3-1}{3}}{\sqrt{\frac{e^2-1}{2}}}.$$

$$\text{proj}_{u_1}(e^{2x}) = \frac{\frac{e^3-1}{3}}{\sqrt{\frac{e^2-1}{2}}} \cdot \frac{e^x}{\sqrt{\frac{e^2-1}{2}}}.$$

$$v_2 = e^{2x} - \text{proj}_{u_1}(e^{2x}).$$

Step 2.3: Compute v_3

$$v_3 = e^{3x} - \text{proj}_{u_1}(e^{3x}) - \text{proj}_{u_2}(e^{3x}),$$

where:

$$\text{proj}_{u_1}(e^{3x}) = \langle e^{3x}, u_1 \rangle u_1, \quad \text{proj}_{u_2}(e^{3x}) = \langle e^{3x}, u_2 \rangle u_2.$$

Step 3: Normalize v_2 and v_3

Normalize v_2 and v_3 to get u_2 and u_3 , respectively:

$$u_2 = \frac{v_2}{\|v_2\|}, \quad u_3 = \frac{v_3}{\|v_3\|}.$$

Result

The orthogonalized set is $\{u_1, u_2, u_3\}$.